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Multimodal Encoding of Čiurlionis' Vocal Music through MEI-TEI Integration and Open LLM Evaluation

Mikalojus Konstantinas Čiurlionis (1875–1911), a Lithuanian composer and painter, developed a body of work in which musical and visual modes of thought are tightly intertwined. Although his piano compositions have recently been made openly accessible through an online platform providing manuscript facsimiles, critical editions, MEI encodings, and interactive analytical tools, comparable computational support for his vocal music remains lacking. Vocal works pose additional challenges due to the need to represent, align, and compare historically evolving textual content alongside musical structure. This paper presents an extension of an existing digital environment for the analysis of Čiurlionis's music that brings his vocal works into focus. The project combines established music-encoding practices with methods from digital text scholarship to represent both music and lyrics in a flexible, historically sensitive way. Multiple versions of song texts are encoded and related directly to musical passages, allowing scholars to examine how changes in words and changes in music interact. In addition, the framework serves as a testing ground for the use of open large language models (LLMs) in musicological practice. We explore how such models can assist in transforming lyric transcriptions into structured scholarly formats and in relating textual variants to musical settings. Several open LLMs are compared with regard to their reliability, consistency, and suitability for historically complex poetic sources. The resulting data model integrates textual and musical materials while remaining compatible with existing encodings of instrumental works. On this basis, we develop tools for searching, comparing, and analysing the relationship between text and music across different versions of a work. By combining close attention to Čiurlionis's vocal repertoire with an evaluation of emerging digital methods, this study offers a general framework for the digital study of vocal music and reflects on the role of AI-based tools in musicological research. The presented research is granted by Lithuanian Research Council (S-LISm-25-7).

Bio:

Jūratė Janutėnaitė-Bogdanienė, PhD, is an associate professor at Vilnius Gediminas Technical University. She has extensive experience in R&D project development and systems engineering, particularly in interdisciplinary research combining information technology, mathematics, and mechanical engineering. She has a strong background in machine learning, software and system development, as well as substantial professional experience as a full-stack Software Engineer. Her research interests include artificial intelligence, mathematical modelling, statistical data analysis, big data, and digital humanities, with a focus on computational methods and digital systems for cultural, musical, and textual heritage studies.

Darius Kučinskas, PhD, is an associate professor at Kaunas University of Technology. He is a Lithuanian musicologist and researcher whose scholarly work focuses on M. K. Čiurlionis studies, music editing and publishing, and the preservation of musical heritage. He has contributed extensively to the fields of digital musicology and historical music media research. Kučinskas is the author and editor of several significant academic publications, including *Music and Technologies* (2014), *Ethnic Piano Rolls in the United States* (2021), and *Chronological Catalogue of the Music of Mikalojus Konstantinas Čiurlionis* (2025), as well as numerous articles on Lithuanian music culture, the history of music publishing, and analytical interpretations of Čiurlionis's compositions. His research has been presented at international conferences and world congresses and published in Lithuanian and international peer-reviewed scholarly journals.